BIOT 6900 · MODULE 2 · PART 2

Detailed Instructions — Getting the Real CPTAC Data Into Your Notebook

Follow these in order. Every step ends with a checkpoint so you know it worked.

This is the step-by-step version. If you're comfortable with Jupyter, the one-page **Quick Start** card is enough. If you want every click spelled out, use this. Part 2 is **not graded** — we do it together in class, and it should take about ten minutes.

Before you start

- You have both notebooks saved in one folder (your "project folder"): `BIOT6900_Module2_Part2_LoadRealCPTAC.ipynb` and `BIOT6900_Module2_Starter.ipynb`.
- Your `biot6900` environment from Module 1 works.
- You have an internet connection and about 10 minutes.

> ✓ **Checkpoint**
> You can open a terminal and typing conda activate biot6900 gives you no error.

Step 1 · Download the three files from LinkedOmics

1. Open a web browser and go to:

```
https://www.linkedomics.org/data_download/CPTAC-BRCA/
```

2. You'll see a table of datasets. Find these three rows and click **Download** on each one (leave the phospho-, acetyl-, and CNV rows alone):

Row on the page	File you'll get
RNAseq (Gene level, Tumor)	`HS_CPTAC_BRCA_2018_RNA_GENE.cct`
Proteome (Gene level, TMT Log Ratio, Tumor)	`HS_CPTAC_BRCA_2018_Proteome_Ratio_Norm_gene_Median.cct`
Mutation (Gene level, Tumor)	`HS_CPTAC_BRCA_2018_MUT_GENE.cbt`

3. **Do not rename or open them.** They usually save to your **Downloads** folder.

> ✓ **Checkpoint**
> You have all three files in your Downloads folder, with the exact names above.

Step 2 · Move the files next to the prep notebook

4. Open the folder that contains `BIOT6900_Module2_Part2_LoadRealCPTAC.ipynb` (your project folder).
5. Drag the three files from Downloads into that same folder.

> ✓ **Checkpoint**
> Your project folder now shows the prep notebook, the main notebook, and the three downloaded files side by side.

Step 3 · Open the prep notebook

6. Open a terminal: **Anaconda Prompt** on Windows, or **Terminal** on Mac.
7. Type these lines, one at a time (replace the path with your own project folder):

```
conda activate biot6900
cd  C:\Users\you\biot6900     # Windows example
cd  ~/biot6900                  # Mac example
jupyter notebook
```

8. A browser tab opens listing your files. Click `BIOT6900_Module2_Part2_LoadRealCPTAC.ipynb` to open it.

> ✓ **Checkpoint**
> The prep notebook is open in your browser, and you can see the file names from Step 1 listed in the file view.

Step 4 · Run the prep notebook

9. In the menu, click `Kernel → Restart & Run All`. If it asks you to confirm, say yes.
10. Wait a few seconds. You should see output like this (numbers may differ slightly):

```
RNA     -> (23121, 122) (genes x samples)
Protein -> (12621, 122) (genes x samples)
Mutation-> 10940 genes (collapsed to per-gene frequency)
cptac_brca_rna.tsv        shape=(23121, 122)  missing=0.3%  TP53 present: True
cptac_brca_protein.tsv    shape=(12621, 122)  missing=...   TP53 present: True
cptac_brca_mutation.tsv   shape=(10940, 1)    missing=0.0%  TP53 present: True

RNA/protein shared tumor samples: 122  (should be ~122)
All three files are now in the data/ folder. You're done here.
```

> **What just happened**
> The notebook oriented each file so genes are the rows, turned the yes/no mutation table into a per-gene frequency, and saved three ready-to-use files into a new `data/` folder. You didn't have to touch any of it.

> ✓ **Checkpoint**
> You see “TP53 present: True” on all three lines, “shared tumor samples: 122,” and no red error message.

Step 5 · Run the main notebook on the real data

11. Back in the browser file list, open `BIOT6900_Module2_Starter.ipynb`.
12. Click `Kernel → Restart & Run All` again.
13. Near the top of Part 1 you should see it print `Loaded CPTAC files from data/.` — **not** the “SYNTHETIC demo data” warning. That confirms it's using your real data.
14. Read the Part 1 output: the RNA–protein correlation median should be modest (around 0.4–0.5), and breast-cancer genes like **TP53, ERBB2, GATA3, PIK3CA** should appear near the top of the ranked list.

> ✓ **Checkpoint**
> Part 1 prints “Loaded CPTAC files from data/.” and the ranked table is full of recognizable breast-cancer genes.

What you should see, and why

The pipeline behaves exactly as it did on the synthetic data: a modest, gene-dependent RNA–protein correlation, and known breast drivers rising to the top. That's the whole point of Part 2 — the **method** is what carries over to real data, not the specific numbers.

> **One honest wrinkle**
> This cohort is all tumor — there are no matched normal samples in these files, so the per-layer signal is overall abundance rather than a tumor-vs-normal difference. And real proteome data has some missing values, so the correlation is computed over the genes with complete measurements. Both are expected.

If something goes wrong

What you see	What to do
conda: command not found	Your environment isn't set up. Open Anaconda Prompt (Windows) specifically, or on Mac run the terminal where conda is installed. Come to office hours if it persists.
jupyter: command not found	You didn't activate the environment. Run `conda activate biot6900` first, then `jupyter notebook`.
Part 1 still says "SYNTHETIC demo data"	The three `.tsv` files aren't in `data/`. Re-run the prep notebook (Step 4) and confirm its last line said the files were written.
FileNotFoundError in the prep notebook	The downloaded files aren't in the same folder as the prep notebook, or a name got changed. Move them next to the notebook and keep the original names.
You can't find the downloads	Check your browser's Downloads folder (or the download bar at the bottom of the browser). Search your computer for "HS_CPTAC_BRCA".
No internet, or download is slow	You can also let the prep notebook fetch the files: open its config cell, set `AUTO_DOWNLOAD = True`, and Run All. This needs a working connection.
A red error you don't recognize	Copy the last few red lines and email them, or bring your laptop to office hours. Don't retype code — the notebook does everything for you.

Getting help

- **Office hours** (weekly, online via Teams) — the fastest fix. Bring your laptop.
- **Email** — include your name, the class title, and the exact red error text or a screenshot. Allow up to 48 hours.